

Regime Detection and Contagion in High-Frequency Limit Order Books: A Temporal Wasserstein Approach

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High-frequency limit order books exhibit regime-dependent dynamics that correlation-based methods cannot fully capture: market microstructure stress propagates through distributional shifts in tail behavior and asymmetry invisible to linear measures. This paper introduces a hierarchical framework combining four innovations. Median Absolute Deviation normalization extracts robust features at sub-second frequencies. Temporal Wasserstein distance detects distributional ruptures within each asset, producing regime signals sensitive to changes in tail shape—a more targeted approach than the cross-sectional formulation used in existing work. A two-level hidden Markov model architecture (per-asset local models feeding a sector Meta-HMM) detects coordinated regime shifts while filtering noise. Transfer Entropy on regime posterior probabilities maps directed financial contagion and identifies which asset initiates sector-wide regime transitions. Applied to LOBSTER high-frequency data for five technology stocks, the framework reveals heterogeneous local regime structures and meaningful global synchronisation. Lead-lag topology inverts between calm and stressed regimes, and directed contagion links concentrate on a primary information hub. An intraday event study validates the Wasserstein detection signal and confirms active cross-sectional reallocation.

Keywords: limit order book; regime detection; wasserstein distance; hidden markov model; financial contagion; market microstructure

JEL Classification: C32, C45, G12, G14

1. Introduction

The increasing fragmentation and algorithmic complexity of modern financial markets have transformed the dynamics of liquidity provision and price discovery. In this context, the study of limit order book (LOB) microstructure has emerged as a critical field, particularly for understanding the mechanisms of contagion and regime-dependent stress propagation across assets. A fundamental challenge is measuring distributional stress in a way that captures the full richness of non-linear dependencies essential for detecting stress regimes. Traditional correlation-based measures prove inadequate: they capture only linear, first-moment relationships and fail to detect the complex distributional changes—shifts in tail behavior, asymmetry, and higher-order moments—that characterize stress propagation in high-frequency markets.

This paper introduces a temporal Wasserstein framework for regime detection and contagion analysis in high-frequency LOB data. Wasserstein distance from optimal transport theory offers a

principled solution: unlike correlation matrices that reduce distributional information to a single linear coefficient, it compares entire probability distributions by computing the minimal transport cost. Critically, I adapt Wasserstein distance to a *temporal* setting by comparing each metric’s distribution *before* versus *after* each time point within a single ticker, producing a stress signal that spikes precisely at distributional ruptures. This is a more targeted signal for regime transitions than the cross-sectional pairwise approach used in existing work (Horvath *et al.* 2021), which conflates persistent cross-asset differences with genuine transitions. For robustness in noisy microstructure data, I use Median Absolute Deviation (MAD) normalization—a non-parametric approach resilient to the extreme outliers (flash quotes, quote stuffing) that destabilize parametric volatility models at sub-second frequencies.

Traditional approaches to regime detection in LOB data suffer from three interrelated limitations this framework directly addresses. First, univariate HMM models (Slupinski 2020, Wisebourn 2011, Nyström 2017) capture latent states within individual assets but overlook the spatial interdependencies between assets crucial in modern fragmented markets. Second, multi-asset LOB models (Kercheval and Zhang 2024, Zhou *et al.* 2025) capture spatial dependencies but rely on correlation-based measures that assume static, linear relationships—assumptions invalidated by the non-linear, regime-dependent stress propagation observed during extreme market events (Cont *et al.* 2014). Third, parametric filtering introduces convergence issues and distributional assumptions that can be problematic at high frequency, blurring genuine regime changes with filtering artifacts.

My framework addresses these limitations through four steps:

- (i) **MAD normalization** of LOB metrics (micro-price returns, order book imbalance, order flow imbalance) to extract robust standardised innovations without imposing distributional assumptions.
- (ii) **Temporal 1-Wasserstein distances** between before-and-after distributions of MAD-normalized metrics, yielding per-ticker rupture signals that capture non-linear, tail, and asymmetric stress patterns invisible to correlations.
- (iii) **A hierarchical HMM architecture:** per-ticker local HMMs with individually optimized parameters extract posterior state probabilities, which a Meta-HMM aggregates into sector-wide regimes. A parallel Direct Global HMM serves as a flat benchmark.
- (iv) **Transfer Entropy on regime posterior probabilities** to identify directed causal relationships and the ‘Patient Zero’ of contagion—who initiates regime transitions, not merely who moves prices first.

Applied to five technology stocks (June 21, 2012), the framework answers a simple question: does one sector behave as one market, or as several micro-markets that occasionally align? The evidence points to a mixed, hierarchical structure. Each ticker follows its own regime geometry; two conceptually different global constructions detect broadly similar sector-wide states (Adjusted Rand Index = 0.371); synchronisation remains low on this non-crisis day; and directed Transfer Entropy links are concentrated rather than uniform. Lead-lag topology inverts between calm and stressed regimes, and an intraday order-flow event reveals coordinated cross-sectional reallocation.

The remainder of this paper is structured as follows. Section 2 reviews the literature. Section 3 details the methodology. Section 4 presents empirical results. Section 5 discusses implications, and Section 6 concludes.

2. Literature Review

Rather than exhaustively covering the microstructure literature, I focus on four axes most relevant to this study: univariate LOB modeling with state-space techniques, multi-asset contagion through spatial dependencies, regime-switching frameworks, and—most critically—Wasserstein distance and information-theoretic measures applied to financial time series.

The foundational work on univariate LOB modeling using Hidden Markov Models established the groundwork for understanding latent market states. Slupinski (2020) demonstrated how HMMs model liquidity regimes by capturing hidden states driving order book imbalances and queue dynamics. Wisebourt (2011) extended this with hierarchical HMM structures to predict returns based on LOB features, while Nyström (2017) implemented non-linear HMM variants for real-time microstructure prediction. All three remain confined to single-asset analysis, overlooking the spatial interdependencies crucial in modern fragmented markets.

The emergence of multi-asset LOB analysis addressed this gap. Kercheval and Zhang (2024) introduced attention-based networks using graph structures to model cross-asset interactions, while Zhou *et al.* (2025) developed transformer-based architectures, explicitly acknowledging non-stationary spatial correlations as a major limitation. Both approaches rely on correlation-based similarity measures that capture only linear, first-moment relationships, missing distributional changes in tails, asymmetry, or higher-order moments.

The regime-switching literature provides the theoretical foundation. Hamilton (1989) established Markov-switching models for business cycle analysis, later adapted by Agnolucci (2019) and Zheng and Li (2020) to financial time series. However, these studies focus on aggregated return or volatility measures and rely on moment-based statistics that overlook the full distributional information distinguishing genuine stress regimes from elevated-but-similar volatility.

Wasserstein distance, originating in optimal transport theory (Villani 2009), offers a principled solution. Peyré and Cuturi (2019) provides efficient algorithms for large-scale applications, and Fournié *et al.* (2019) establishes theoretical foundations for financial time series. Horvath *et al.* (2021) demonstrate that Wasserstein-based regime detection successfully identifies bull, bear, and crisis periods in equity indices by capturing distributional shifts invisible to correlation matrices. Table F1 summarizes the four key extensions of my framework relative to Horvath *et al.* (2021).

[Table F1 near here]

Each extension is consequential. Sub-second resolution reveals microstructure phenomena invisible at daily frequency. MAD normalization avoids convergence failures common with parametric models at high frequency. The temporal Wasserstein formulation provides a self-contained per-ticker rupture signal without requiring cross-asset alignment. The HMM’s enforced persistence captures the economic reality that microstructure regimes exhibit substantial inertia.

Transfer Entropy (Schreiber 2000) extends mutual information to measure directed information flow between time series, providing a non-parametric measure of causal influence. My framework applies Transfer Entropy to regime posterior probabilities rather than raw returns, enabling detection of who initiates regime transitions—a question correlation-based lead-lag analysis cannot answer.

3. Methodology

The pipeline proceeds in five stages: data loading and LOB metric construction, MAD normalization, temporal Wasserstein feature extraction, hierarchical HMM estimation with per-ticker optimization, and contagion analysis. Figure F summarizes the architecture.

[Figure F near here]

3.1. Data and LOB metrics

I utilize LOBSTER (Limit Order Book System – The Efficient Reconstructor) data (LOBSTER Data GmbH 2019) for five technology stocks (AAPL, INTC, GOOG, AMZN, MSFT) on June 21, 2012 (9:30–16:00 ET). Data are synchronized to a common 500 ms grid using forward-fill interpolation, yielding approximately 46,400 observations per asset.

I compute three metrics capturing complementary aspects of market dynamics. The *micro-price* serves as the primary measure of fair value, calculated as a volume-weighted average across the

first $n = 5$ levels:

$$M_{t,n} = \frac{\sum_{i=1}^n (P_{i,t}^{bid} \cdot Q_{i,t}^{ask} + P_{i,t}^{ask} \cdot Q_{i,t}^{bid})}{\sum_{i=1}^n (Q_{i,t}^{bid} + Q_{i,t}^{ask})} \quad (1)$$

The *Order Book Imbalance* (OBI) measures static relative supply-demand pressure:

$$\text{OBI}_{t,n} = \frac{\sum_{i=1}^n Q_{i,t}^{bid} - \sum_{i=1}^n Q_{i,t}^{ask}}{\sum_{i=1}^n Q_{i,t}^{bid} + \sum_{i=1}^n Q_{i,t}^{ask}} \quad (2)$$

The *Multi-Level Order Flow Imbalance* (OFI) tracks the dynamic net liquidity flow, extending the framework of Cont *et al.* (2014) to aggregate changes across five levels:

$$\text{OFI}_{t,n} = \sum_{i=1}^n (e_{i,t}^{bid} - e_{i,t}^{ask}) \quad (3)$$

where the bid-side contribution $e_{i,t}^{bid}$ at each level accounts for price improvements, size changes at constant price, or price retreats.

3.2. Robust MAD normalization

A critical departure from standard parametric filtering is the adoption of Median Absolute Deviation (MAD) normalization. High-frequency microstructure data exhibit extreme outliers—flash quotes, quote stuffing, transient liquidity vacuums—that destabilize parametric volatility models. For a rolling window of size w , I compute the robust z-score:

$$Z_t = \frac{X_t - \tilde{X}_t}{1.4826 \cdot \text{MAD}_t + \epsilon} \quad (4)$$

where $\tilde{X}_t = \text{median}(X_{t-w}, \dots, X_t)$, $\text{MAD}_t = \text{median}(|X_{t-w} - \tilde{X}_t|, \dots, |X_t - \tilde{X}_t|)$, and $\epsilon = 10^{-9}$. The constant 1.4826 ensures consistency with the standard deviation under normality. Three properties suit this to high-frequency LOB data: breakdown point 0.5 (up to 50% outliers do not affect the estimate); no iterative optimization; and no distributional assumption on the innovation process. The window size w is optimized per ticker, ranging from 50 to 150 observations.

3.3. Temporal Wasserstein distance

Why not correlation. Standard correlation-based measures reduce cross-sectional dependencies to a single linear coefficient, discarding all distributional shape beyond means and variances. For high-frequency LOB data, this is inadequate on three counts: regime changes manifest as shifts in skewness, kurtosis, and tail behavior that correlations cannot detect by construction; dependencies are highly non-linear (threshold effects, asymmetric propagation); and extreme conditional heteroskedasticity renders correlation unstable across time.

Wasserstein distance. The 1-Wasserstein distance between two probability distributions P and Q on $\mathbb{R}$ is:

$$W_1(P, Q) = \inf_{\gamma \in \Gamma(P, Q)} \mathbb{E}_{(X, Y) \sim \gamma} [|X - Y|] = \int_0^1 |F_P^{-1}(u) - F_Q^{-1}(u)| du$$

where $\Gamma(P, Q)$ denotes the set of all joint distributions with marginals P and Q , and F_P^{-1}, F_Q^{-1} are the quantile functions. Four properties make it well-suited for distributional stress measurement: it is distribution-free (no normality, moment, or linearity assumptions); it captures full distributional information by comparing quantile functions at all levels $u \in [0, 1]$; it is geometrically interpretable as the minimum transport cost to transform P into Q ; and it satisfies the metric axioms, ensuring regimes correspond to well-defined clusters in distribution space. I select $p = 1$ over higher-order variants because it weights all quantile levels equally (avoiding over-weighting of tail extremes that may reflect idiosyncratic noise), admits a closed-form $O(n \log n)$ solution via sorted quantiles, and measures marginal distributional stress independently of temporal correlation structure, which the HMM captures subsequently.

From cross-sectional to temporal. The existing literature (Horvath *et al.* 2021) applies Wasserstein distance cross-sectionally, computing $W_1(\hat{P}_{i,t}, \hat{P}_{j,t})$ between assets i and j within a rolling window. This conflates genuine within-asset distributional regime changes with persistent cross-asset differences. My temporal formulation resolves this by comparing each metric’s distribution *before* versus *after* each time point *within* the same ticker:

$$W_t^{(i,m)} = W_1\left(\{Z_{t-w}^{(i,m)}, \dots, Z_t^{(i,m)}\}, \{Z_t^{(i,m)}, \dots, Z_{t+w}^{(i,m)}\}\right) \quad (5)$$

where $Z^{(i,m)}$ is the MAD-normalized series for asset i and metric m . Three structural consequences follow: $W_t^{(i,m)}$ is small when the distribution is stationary and spikes at genuine distributional ruptures; each series is self-contained per ticker, enabling per-ticker HMM fitting without cross-asset alignment; and MAD normalization ensures spikes reflect genuine shape changes (skewness, kurtosis, modality) rather than level or scale shifts.

3.4. Hierarchical HMM architecture

The core architecture is a two-level HMM that resolves three fundamental problems of flat multi-asset HMMs: label switching, noise filtering, and contagion detection.

At the first level, I fit a Gaussian HMM to the temporal Wasserstein features for each ticker i independently:

$$\mathbf{W}_t^{(i)} | S_t^{(i)} = k \sim \mathcal{N}(\boldsymbol{\mu}_k^{(i)}, \boldsymbol{\Sigma}_k^{(i)})$$

The covariance structure defaults to diagonal; it switches to full covariance if the maximum absolute correlation between features exceeds $\tau = 0.70$. To mitigate convergence to suboptimal local maxima, I run $n_{\text{init}} = 5$ independent restarts per ticker, each with up to 200 EM iterations, selecting the best restart by observed-data log-likelihood evaluated via the forward algorithm. After EM estimation, I enforce temporal persistence:

$$\mathbf{A}_{jk}^{(i)} = \begin{cases} \pi^{(i)} & \text{if } j = k \\ \frac{1-\pi^{(i)}}{K-1} & \text{if } j \neq k \end{cases} \quad (6)$$

and apply post-estimation smoothing via a majority filter of window $s^{(i)}$. Beyond Viterbi-decoded state labels, I extract posterior state probabilities via the forward-backward algorithm:

$$\mathbf{p}_t^{(i)} = \left(P(S_t^{(i)} = 0 | \mathbf{W}_{1:T}^{(i)}), \dots, P(S_t^{(i)} = K-1 | \mathbf{W}_{1:T}^{(i)})\right) \quad (7)$$

These continuous probabilities serve as input to the Meta-HMM, preserving uncertainty and resolving label switching.

At the second level, the Meta-HMM observes the concatenated local state probabilities: $\mathbf{X}_t^{\text{meta}} = [\mathbf{p}_t^{(1)}, \dots, \mathbf{p}_t^{(N)}] \in \mathbb{R}^{N \times K}$. It is a Gaussian HMM with K_G global regimes, applying the same persistence and smoothing procedure. The hierarchical design filters noise through a consensus mechanism and directly detects contagion as correlated changes in local state probabilities. As a benchmark, I fit a Direct Global HMM on concatenated temporal Wasserstein features: $\mathbf{X}_t^{\text{direct}} = [\mathbf{W}_t^{(1)}, \dots, \mathbf{W}_t^{(N)}] \in \mathbb{R}^{N \times 3}$.

3.5. Per-ticker parameter optimization

Rather than imposing a single set of hyperparameters, I search over a grid for each ticker independently, varying the MAD window ($w_{\text{MAD}} \in \{50, 100, 150\}$), the Wasserstein window ($w_{\text{Wass}} \in \{50, 100, 150\}$), the local persistence ($\pi \in \{0.85, 0.90, 0.95\}$), and the smoothing window ($s \in \{10, 20, 30\}$), with $K = 3$ regimes fixed. Each configuration is evaluated by:

$$\text{Score} = \text{ARI}(\text{HMM}, \text{K-means}) - \lambda \cdot \text{MMD}_{\text{penalty}} \quad (8)$$

where the Adjusted Rand Index measures agreement with non-parametric K-means clustering, the MMD penalty discourages configurations with inconsistent distributional separation, and $\lambda = 0.10$.

3.6. Contagion metrics

Transfer Entropy measures directed information flow from a source to a target asset:

$$\text{TE}(X \rightarrow Y) = \sum_{y_{t+1}, y_t, x_t} p(y_{t+1}, y_t, x_t) \log \frac{p(y_{t+1}|y_t, x_t)}{p(y_{t+1}|y_t)} \quad (9)$$

where x_t and y_t are discretized versions of the source and target stress probabilities (sum of non-calm regime probabilities), using $k = 2$ lags and 10 bins. I compute the full $N \times N$ Transfer Entropy matrix and combine mean outgoing Transfer Entropy with Meta-HMM synchronisation leadership scores to rank tickers by their potential to initiate sector-wide regime shifts.

To capture state dependence in contagion strength, lead-lag correlations are estimated separately within stress quantiles (Q10, Q50, Q90) of the temporal Wasserstein stress signal, isolating whether directional dependencies are stable across market conditions or amplify in tail-stress episodes.

3.7. Robustness framework

I validate through four complementary diagnostics: Adjusted Rand Index between HMM and K-means clustering (genuine structure vs parametric artifacts); Maximum Mean Discrepancy between regime-conditional distributions (distributional separation quality); Meta-HMM vs Direct Global HMM state agreement (hierarchical vs flat detection); and posterior entropy (confident vs diffuse regime assignments, with near-zero entropy indicating deterministic classification).

4. Results

4.1. Per-ticker optimization and local regimes

Optimised local HMM parameters (Table B1) reveal substantial heterogeneity across assets. AAPL and AMZN favour short MAD and Wasserstein windows (50 observations), suggesting their microstructure distributions shift at relatively short time scales. GOOG requires a longer MAD window (100 observations) and the longest Wasserstein window (150), indicating that distributional

ruptures are more gradual. INTC occupies an intermediate position with short MAD window (50) but longer Wasserstein window (100) and the highest persistence ($\pi = 0.95$). MSFT is characterized by the longest MAD window (150) and the shortest Wasserstein window (50). This cross-asset heterogeneity validates per-ticker optimization: a single configuration applied uniformly would systematically misfit several tickers.

[Table B1 near here]

Regime labels (Calm, Intermediate, Stressed) are assigned post-estimation by a composite stress proxy on the *original* LOB features:

$$\tilde{\sigma}_t = \text{norm}_{99}(|\Delta p_t/p_t|) + \text{norm}_{99}(|\text{OBI}_t|) + \text{norm}_{99}(|\text{OFI}_t|) \quad (10)$$

where norm_{99} denotes robust min-max scaling clipped to the 1st–99th percentile. The per-regime mean $\tilde{\sigma}_r$ determines the label ordering. Detailed descriptive statistics are in Appendix A.

Figures F–F display the Viterbi regime sequences. Ticker-level patterns are strikingly heterogeneous:

[Figures F–F near here]

AAPL: Dominant regime Intermediate (56.0%), with Stressed (34.6%) and Calm (9.4%). Return volatility is the primary discriminant: the Stressed regime carries substantially higher σ_r (0.73 bp) and kurtosis (62.1) than Calm (0.44 bp, kurtosis 11.3), indicating sparse large return spikes rather than sustained elevated volatility.

INTC: Highly concentrated—Calm (80.6%) dominates, with Intermediate (14.2%) and Stressed (5.2%) as smaller regimes. Return volatility is non-monotonic, so OBI drives the stress ordering: the Stressed regime carries the highest $|\overline{\text{OBI}}|$ (0.12). Very high kurtosis across all regimes (94–135) indicates uniformly fat-tailed returns, a signature of thinner market structure.

GOOG: Atypical structure—a single dominant Stressed state (95.1%) covers nearly the entire trading day. This reflects GOOG’s characteristically high intraday microstructure intensity—a highly liquid mega-cap where typical conditions involve substantial price discovery and book imbalance—not persistent market distress in an absolute sense.

AMZN: Dominant Intermediate (65.6%), with Calm (23.2%) and Stressed (11.2%). The Stressed regime achieves its label through the highest $|\overline{\text{OFI}}|$ (241 vs 181 for Intermediate), consistent with institutional order-slicing rather than large price moves.

MSFT: Dominant Intermediate (67.4%), with Stressed (30.2%) and Calm (2.4%). Return volatility increases monotonically from 0.40 bp (Calm) to 0.69 bp (Stressed), with OBI and OFI confirming coherent stress ordering.

4.2. Global regimes and synchronisation

The Meta-HMM and Direct Global HMM identify sector-wide regimes from fundamentally different inputs yet produce sequences with substantial state agreement (75.7%, Table E1). The ARI between the two models is 0.371, indicating meaningful structural agreement. Both models produce near-deterministic assignments (mean entropy 0.0006 and 0.0004 respectively), confirming that disagreements arise from regime boundary placement rather than posterior uncertainty. The Direct model achieves stronger distributional separation ($\text{MMD}_{\text{OFI}} = 0.726$ vs 0.710; $\text{MMD}_{\text{OBI}} = 0.769$ vs 0.602), while the Meta-HMM remains interpretable through its hierarchical construction.

Synchronisation between each ticker’s local HMM and the Meta-HMM is reported in Table C1, across 51 global transitions. Co-transition synchronisation is sparse: AAPL and INTC have 0 co-transitions, GOOG has 7 (13.7%), and AMZN and MSFT each have 1 (2.0%). The overall picture is sparse synchronisation with limited but non-zero anticipatory structure—consistent with a non-crisis day where idiosyncratic microstructure dynamics dominate.

[Tables C1 and E1 near here]

4.3. Stress decomposition and lead-lag dynamics

Figure F presents the ticker-specific Wasserstein stress decomposition with Meta-HMM regime overlays. Periods flagged as elevated-stress by the Meta-HMM correspond to visible co-elevations across multiple tickers in the OFI channel, validating that the Meta-HMM identifies economically coherent periods of coordinated distributional change.

[Figure F near here]

The lead-lag analysis reveals several key patterns:

State-dependent topology inversion (key finding). In the low-stress regime (Q10), GOOG leads the sector—GOOG leads AAPL by +2 s ($r = 0.393$), MSFT by +8 s ($r = 0.278$), and INTC ($r = 0.160$). Under high stress (Q90), the network inverts: INTC becomes the dominant leader, anticipating AAPL by 7.5 s ($r = 0.526$, the strongest pairwise correlation across all quantiles and pairs), MSFT near-instantaneously ($r = 0.387$), and AAPL at +10 s ($r = 0.381$). AMZN also leads AAPL during stress ($r = 0.372$, +2.5 s).

Non-linear stress amplification. Q90 correlations systematically exceed Q10 across all pairs, confirming that microstructure dependencies intensify under stress in a way that static correlation matrices—which conflate distributional change with heteroskedasticity—cannot detect.

Aggregate unconditional hierarchy. INTC–MSFT exhibits the strongest unconditional dependency ($r = 0.253$ at +10 s, with INTC leading). AAPL–AMZN is second ($r = 0.222$ at +7 s). Two pairs show negligible dependency: AAPL–GOOG ($r = -0.010$) and INTC–AMZN ($r = 0.007$), confirming that not all within-sector pairs are coupled at microstructure frequencies.

OFI clustering. OFI exhibits strong autocorrelation ($\rho > 0.8$ within ± 5 s), indicating order flow toxicity arrives in persistent bursts consistent with algorithmic order-slicing. The OBI channel yields the single strongest pairwise correlation in the dataset: GOOG→AAPL at Q90 ($r = 0.54$, −4 s).

[Figures F and F near here]

4.4. Transfer Entropy and contagion structure

The strongest Transfer Entropy links are AMZN→MSFT (TE = 0.0035), INTC→MSFT (0.0022), AAPL→MSFT (0.0017), GOOG→MSFT (0.0013), and MSFT→INTC (0.0013). MSFT emerges as the dominant information recipient in the top Transfer Entropy channels. Combining the Transfer Entropy matrix with synchronisation leadership scores reveals a layered contagion structure: GOOG is the most synchronized ticker with the global regime (7/51 co-transitions) yet AAPL and INTC show zero co-transitions; the largest Transfer Entropy channels converge on MSFT as the principal information recipient. This heterogeneous picture—sparse co-transitions, concentrated directed TE links—is invisible to symmetric correlation analysis.

[Table D1 near here]

4.5. Event study: GOOG OFI spike as in-sample validation

This section uses the episode at $t \approx 15,000$ ($\sim 11:33$ AM) not as a standalone illustration, but as an in-sample validation of three framework outputs: the temporal Wasserstein detection signal, the Meta-HMM regime assignment, and the Transfer Entropy topology.

Step 1 — Framework detection. The event is identified by the framework’s primary output: the temporal Wasserstein OFI stress signal $W_t^{(\text{GOOG}, \text{OFI})}$ reaches an elevated value at $t \approx 15,000$, visible as a spike in the bottom panel of Figure F (GOOG line, OFI row). This spike satisfies the temporal formulation’s detection criterion: the OFI distribution over the post-window diverges significantly from the pre-window distribution, signalling a genuine distributional rupture in GOOG’s order flow.

Step 2 — Regime context from the Meta-HMM. Figure F simultaneously reveals the regime context via its background shading. Consistent with GOOG having the highest co-transition synchronisation rate (7/51 global transitions, Table C1), this episode coincides with one of GOOG’s rare local regime transitions—a sharp change in the posterior probability vector $\mathbf{p}_t^{(\text{GOOG})}$ that propagates through the Meta-HMM input and triggers a global regime transition. The episode illustrates the full vertical integration of the hierarchical architecture: ticker-level distributional rupture $\rightarrow$ local regime shift $\rightarrow$ sector-wide Meta-HMM transition.

Step 3 — Transfer Entropy prediction. The Transfer Entropy network provides a structural prediction about the absorption pattern. All four top directional channels in the sector converge on MSFT as the unique primary information recipient. If the directed information topology reflects underlying capital flow dynamics, MSFT should absorb the dominant fraction of cross-sectional reallocation following a stress event in any other ticker.

Step 4 — Observation and validation. Figure F confirms this structural prediction. GOOG exhibits $\Delta\text{OFI} \approx -55$ at $t \approx 15,000$, while MSFT, AMZN, AAPL, and INTC show positive OFI changes totalling approximately +450. The absorption hierarchy is MSFT (+218, 48.5%) > AMZN (+112, 24.9%) > AAPL (+68, 15.2%) > INTC (+52, 11.5%). MSFT’s dominant absorption role validates, in-sample, the Transfer Entropy topology’s designation of MSFT as the sector’s primary information hub.

[Figure F near here]

Three caveats apply. First, the Transfer Entropy was estimated on the full trading day including this episode: the match is in-sample rather than a genuine out-of-sample prediction. Second, the Transfer Entropy ordering across non-MSFT tickers does not map cleanly to the absorption ordering, reflecting the fact that Transfer Entropy captures general directed information flow over the full session rather than specifically GOOG-originated flows. Third, the contemporaneous cross-sectional response remains consistent with a common latent shock or shared algorithmic response.

4.6. Robustness diagnostics

The Direct Global HMM aligns better with the non-parametric K-means benchmark ($\text{ARI} = 0.259$) than the Meta-HMM ($\text{ARI} = 0.212$), while meta-vs-direct ARI is 0.371. MMD diagnostics show stronger separation for the Direct model (OBI: 0.769 vs 0.602; OFI: 0.726 vs 0.710 for Direct vs Meta). Both models assign near-deterministic regimes (mean entropy 0.0006 and 0.0004), and their state agreement is high (75.7%), indicating that differences arise from boundary placement rather than probabilistic ambiguity. Full diagnostic tables are in Appendix E.

4.7. Summary

The combined evidence describes a coherent sector-level story. Directed information flows concentrate toward MSFT; global synchronisation remains sparse on this non-crisis day (most tickers at 0–1 co-transitions out of 51); lead-lag topology is strongly state-dependent, with materially stronger dependencies under stressed quantiles; and a GOOG-specific OFI shock triggers rapid cross-ticker reallocation. The contagion mechanism is directional, state-dependent, and heterogeneous across tickers—none of which is detectable from unconditional correlation analysis.

5. Discussion

5.1. *Methodological contributions*

The temporal Wasserstein formulation is the central methodological contribution. By detecting *when* a distributional change occurs within each ticker rather than measuring how different two assets are cross-sectionally, it provides a rupture signal that is self-contained per asset, resolves the confound between persistent cross-asset differences and genuine transitions, and naturally feeds a hierarchical architecture. The MAD normalization ensures this rupture signal reflects genuine shape changes rather than local level or scale shifts.

The hierarchical architecture contributes three distinct mechanisms absent from flat HMMs: it resolves label switching by operating on posterior probabilities rather than discrete labels; it filters noise through a consensus mechanism; and it enables direct contagion detection as correlated changes in local state probabilities. The comparison between Meta-HMM and Direct Global HMM shows both approaches detect broadly similar structures ($\approx 72\%$ state agreement) with complementary strengths—the Meta-HMM edges out on OBI separation, the Direct model on OFI separation—suggesting that combining their signals could improve detection further.

The integration of Transfer Entropy with regime posterior probabilities enables a qualitatively different question: not who moves prices first, but who initiates regime transitions. The layered contagion structure that emerges (anticipatory initiator, direct co-transition partner, secondary amplifiers, net followers) is invisible to symmetric correlation analysis.

5.2. *Economic implications*

The heterogeneity of local regime structures is a key empirical finding. Rather than a single regime characterising the technology sector, each asset exhibits its own microstructure signature: GOOG spends 95% of the day in a single dominant state while MSFT and AAPL distribute time more evenly. This heterogeneity reflects differences in market-making activity, institutional ownership, and algorithmic trading intensity.

The weak but structured global synchronisation is consistent with a non-crisis day. Co-transition rates remain low (0.0%–13.7%), indicating idiosyncratic dynamics dominate sector-wide co-movement. This provides a useful baseline: on a crisis day, one would expect this sparse structure to compress into broader simultaneous co-transitions as common shocks override idiosyncratic dynamics.

The state-dependent topology inversion—GOOG leading in calm, INTC leading under stress—implies that the effective information hierarchy of the sector is regime-conditional. A risk monitoring system measuring leadership unconditionally would misidentify the early warning signal in stressed environments.

5.3. *Practical applications*

For risk managers, the framework provides real-time monitoring of cross-sectional stress and actionable Patient Zero identification, calibrated to each asset’s microstructure. The 52-second average regime duration provides sufficient lead time for portfolio hedging once a transition is detected.

For high-frequency traders, the OBI and OFI lead-lag patterns under stressed quantiles represent exploitable predictability windows—but strategies must be stress-aware since these edges are substantially weaker during calm periods. MSFT’s consistent role as principal Transfer Entropy recipient creates predictable buy-side pressure during sector stress.

The stabilizing nature of the cross-sectional reallocation documented in the event study argues against overly restrictive high-frequency trading regulations: stress is absorbed via redistribution rather than amplification.

5.4. *Limitations and future directions*

Several limitations should be noted. The analysis covers a single trading day, and extending to multiple days including crisis periods would test the framework’s ability to detect varying contagion intensities. The per-ticker optimization uses a fixed grid rather than adaptive or Bayesian search. The Transfer Entropy computation relies on histogram-based discretization (sensitive to bin count); continuous estimators such as the KSG estimator could improve robustness. The framework currently focuses on within-sector contagion; cross-sector analysis would provide a more complete picture of systemic risk propagation.

6. Conclusion

This paper introduces a hierarchical regime-detection and contagion framework for high-frequency limit order books combining four methodological innovations: robust MAD normalization, temporal Wasserstein distances (before vs after, within each ticker), a two-level HMM architecture (local per-ticker HMMs aggregated by a Meta-HMM), and Transfer Entropy on regime posterior probabilities for causal analysis.

Empirically, the answer to the motivating question—does one sector behave as one market?—is nuanced: heterogeneous locally, partially coherent globally, and hierarchically connected through time-varying directed links. Lead-lag topology inverts between calm and stressed regimes, Transfer Entropy reveals concentrated directed channels, and an intraday event study confirms active portfolio rebalancing with MSFT consistently absorbing the largest share of sector flows. Robustness validation confirms that these findings reflect genuine market structure rather than modeling artifacts.

The framework provides a principled, production-ready foundation for regime-aware microstructure analysis: the distributional sensitivity of Wasserstein distance, the temporal structure of HMMs, and the causal directionality of Transfer Entropy, integrated into a single pipeline applicable to real-time monitoring of cross-sectional stress and portfolio risk management.

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Disclosure of Interest

The author reports there are no competing interests to declare.

Declaration of Generative AI Use

During the preparation of this work, the author used generative AI tools in two capacities. First, Claude (Anthropic) was used to assist with manuscript editing and structural improvements to language and presentation. Second, Claude Code (Anthropic) was used to assist with Python code development and repository management in the production of the analysis pipeline; this includes code that automated the generation of figures and tables. The author reviewed and edited all AI-assisted content, verified all outputs, and takes full responsibility for the integrity of the final manuscript. Generative AI was not used in the research design, data collection, statistical analysis, interpretation of results, or the scientific conclusions drawn from the empirical findings.

Data Availability Statement

The data used in this study are sourced from the LOBSTER database (<https://lobsterdata.com>), available upon academic subscription. The analysis code is available from the author upon reasonable request.

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Appendix A: Regime descriptive statistics

Tables A1–A5 report descriptive statistics broken down by HMM regime and ticker. Reported columns are price return volatility σ_r , mean absolute return $|\bar{r}|$, return excess kurtosis, mean absolute OBI, and mean absolute OFI. Regimes are ordered and labeled by the composite stress proxy of Equation (10). σ_r and $|\bar{r}|$ are in basis points (1 bp = 10^{-4}).

Table A1. Regime descriptive statistics – AAPL.

Regime	Label	N (%)	σ_r (bp)	$ \bar{r} $ (bp)	Kurt	$ \overline{\text{OBI}} $	$ \overline{\text{OFI}} $
R2	Calm	4,358 (9.4%)	0.44	0.22	11.3	0.32	230
R0	Intermediate	25,996 (56.0%)	0.50	0.24	16.3	0.34	242
R1	Stressed	16,046 (34.6%)	0.73	0.37	62.1	0.32	275

Table A2. Regime descriptive statistics – INTC.

Regime	Label	N (%)	σ_r (bp)	$ \bar{r} $ (bp)	Kurt	$ \overline{\text{OBI}} $	$ \overline{\text{OFI}} $
R0	Calm	37,400 (80.6%)	0.60	0.12	93.7	0.09	371
R2	Intermediate	6,582 (14.2%)	0.62	0.10	134.7	0.10	397
R1	Stressed	2,418 (5.2%)	0.40	0.06	101.1	0.12	392

Table A3. Regime descriptive statistics – GOOG.

Regime	Label	N (%)	σ_r (bp)	$ \bar{r} $ (bp)	Kurt	$ \overline{\text{OBI}} $	$ \overline{\text{OFI}} $
R1	Calm	626 (1.3%)	0.37	0.14	21.6	0.23	145
R2	Intermediate	1,669 (3.6%)	0.56	0.19	36.6	0.22	219
R0	Stressed	44,105 (95.1%)	0.68	0.27	27.8	0.31	162

Table A4. Regime descriptive statistics – AMZN.

Regime	Label	N (%)	σ_r (bp)	$ \bar{r} $ (bp)	Kurt	$ \overline{\text{OBI}} $	$ \overline{\text{OFI}} $
R2	Calm	10,776 (23.2%)	0.84	0.33	28.1	0.34	178
R0	Intermediate	30,431 (65.6%)	0.88	0.39	23.4	0.33	181
R1	Stressed	5,193 (11.2%)	0.68	0.26	21.2	0.33	241

Table A5. Regime descriptive statistics – MSFT.

Regime	Label	N (%)	σ_r (bp)	$ \bar{r} $ (bp)	Kurt	$ \overline{\text{OBI}} $	$ \overline{\text{OFI}} $
R2	Calm	1,103 (2.4%)	0.40	0.06	97.4	0.07	391
R0	Intermediate	31,282 (67.4%)	0.60	0.12	56.3	0.08	400
R1	Stressed	14,015 (30.2%)	0.69	0.15	45.1	0.09	440

Appendix B: Local HMM configuration

Table B1. Local HMM optimized parameters

ticker	mad_window	wasserstein_window	local_persistence	local_smoothing	n_regimes
AAPL	50	150	0.8500	20	3
INTC	50	100	0.9500	10	3
GOOG	100	150	0.8500	10	3
AMZN	50	150	0.8500	10	3
MSFT	150	50	0.9000	10	3

Appendix C: Local–global synchronisation

Table C1. Local-to-global synchronization rate

Ticker	Sync Rate	co.transitions	total.transitions_global	leadership_score	leads	lags
AAPL	0.0000	0	51	1.0000	2	0
GOOG	0.1373	7	51	1.0000	7	0
AMZN	0.0196	1	51	1.0000	1	0
MSFT	0.0196	1	51	0.1111	20	16
INTC	0.0000	0	51	0.0000	2	2

Appendix D: Contagion network and inter-ticker lead-lag

Table D1. Transfer Entropy (top)

source	target	te
AMZN	MSFT	0.0035
INTC	MSFT	0.0022
AAPL	MSFT	0.0017
GOOG	MSFT	0.0013
MSFT	INTC	0.0013
MSFT	AMZN	0.0012
AAPL	INTC	0.0010
MSFT	AAPL	0.0009
INTC	AMZN	0.0008
AMZN	AAPL	0.0007

Table D2. Lead-lag between tickers (top results)

quantile	ticker1	ticker2	best_lag_obs	best_lag_seconds	best_corr	best_pval	n_obs
Q10	GOOG	AAPL	4	2.0000	0.3930	0.0000	4640
Q10	GOOG	AMZN	-20	-10.0000	0.3351	0.0000	4640
Q10	GOOG	MSFT	-16	-8.0000	0.2776	0.0000	4640
Q10	INTC	GOOG	-18	-9.0000	0.2619	0.0000	4640
Q10	GOOG	INTC	5	2.5000	0.1597	0.0000	4640
Q10	MSFT	AAPL	0	0.0000	0.1542	0.0000	4640
Q10	MSFT	AMZN	0	0.0000	0.1408	0.0000	4640
Q10	INTC	AMZN	-17	-8.5000	0.1258	0.0000	4640
Q10	AMZN	MSFT	-10	-5.0000	0.0824	0.0000	4640
Q10	MSFT	INTC	-12	-6.0000	0.0780	0.0000	4640
Q10	AAPL	INTC	-19	-9.5000	-0.0761	0.0000	4640
Q10	INTC	MSFT	8	4.0000	-0.0704	0.0000	4640
Q10	AAPL	GOOG	0	0.0000	0.0699	0.0000	4640
Q10	AAPL	AMZN	-17	-8.5000	0.0691	0.0000	4640
Q10	MSFT	GOOG	5	2.5000	-0.0606	0.0000	4640
Q10	AAPL	MSFT	-10	-5.0000	0.0511	0.0005	4640
Q10	AMZN	AAPL	20	10.0000	0.0374	0.0111	4640
Q10	AMZN	GOOG	20	10.0000	0.0292	0.0472	4640
Q50	INTC	AAPL	17	8.5000	0.2944	0.0000	23200
Q50	AMZN	AAPL	20	10.0000	0.2792	0.0000	23200
Q50	MSFT	INTC	-12	-6.0000	0.2301	0.0000	23200
Q50	AAPL	INTC	-20	-10.0000	0.2221	0.0000	23200
Q50	MSFT	AMZN	0	0.0000	0.1758	0.0000	23200
Q50	GOOG	AAPL	20	10.0000	-0.1617	0.0000	23200
Q50	MSFT	AAPL	-11	-5.5000	0.1532	0.0000	23200
Q50	MSFT	GOOG	-20	-10.0000	-0.1255	0.0000	23200
Q50	AMZN	MSFT	20	10.0000	0.1239	0.0000	23200
Q50	AAPL	MSFT	-3	-1.5000	0.0867	0.0000	23200
Q50	AAPL	AMZN	-7	-3.5000	0.0778	0.0000	23200
Q50	AAPL	GOOG	12	6.0000	0.0653	0.0000	23200
Q50	GOOG	MSFT	-20	-10.0000	0.0589	0.0000	23200
Q50	AMZN	GOOG	20	10.0000	-0.0498	0.0000	23200
Q50	GOOG	AMZN	-20	-10.0000	0.0496	0.0000	23200
Q50	AMZN	INTC	-20	-10.0000	0.0438	0.0000	23200
Q50	INTC	MSFT	-20	-10.0000	-0.0323	0.0000	23200
Q50	INTC	AMZN	1	0.5000	0.0243	0.0002	23200
Q50	GOOG	INTC	-2	-1.0000	0.0165	0.0122	23200
Q90	AAPL	INTC	-15	-7.5000	0.5261	0.0000	4640
Q90	MSFT	INTC	-1	-0.5000	0.3869	0.0000	4640
Q90	INTC	AAPL	20	10.0000	0.3806	0.0000	4640
Q90	AMZN	AAPL	5	2.5000	0.3722	0.0000	4640
Q90	MSFT	AAPL	13	6.5000	0.3162	0.0000	4640
Q90	GOOG	AMZN	3	1.5000	0.2634	0.0000	4640
Q90	GOOG	INTC	20	10.0000	0.2549	0.0000	4640
Q90	GOOG	MSFT	-2	-1.0000	0.2159	0.0000	4640
Q90	AAPL	MSFT	-2	-1.0000	0.2129	0.0000	4640
Q90	GOOG	AAPL	-5	-2.5000	0.2060	0.0000	4640
Q90	MSFT	GOOG	20	10.0000	-0.2059	0.0000	4640
Q90	MSFT	AMZN	0	0.0000	0.1620	0.0000	4640
Q90	AAPL	AMZN	20	10.0000	0.1531	0.0000	4640
Q90	INTC	GOOG	20	10.0000	-0.1516	0.0000	4640
Q90	INTC	AMZN	10	5.0000	0.1045	0.0000	4640
Q90	AAPL	GOOG	6	3.0000	-0.0781	0.0000	4640
Q90	AMZN	GOOG	-4	-2.0000	0.0689	0.0000	4640
Q90	INTC	MSFT	20	10.0000	0.0580	0.0001	4640

Appendix E: Robustness diagnostics

Table E1. Unified robustness diagnostics (ARI, MMD, entropy)

Metric	Value
ARI meta vs K-Means	0.2123
ARI direct vs K-Means	0.2587
ARI meta vs direct	0.3711
ARI meta vs local (AAPL)	0.0256
ARI meta vs local (INTC)	0.1328
ARI meta vs local (GOOG)	0.6434
ARI meta vs local (AMZN)	0.1078
ARI meta vs local (MSFT)	0.1388
MMD meta (Price)	0.1742
MMD direct (Price)	0.1879
MMD meta (OFI)	0.7099
MMD direct (OFI)	0.7257
MMD meta (OBI)	0.6024
MMD direct (OBI)	0.7688
Entropy mean (meta)	0.0006
Entropy mean (direct)	0.0004
State sync (meta vs direct)	0.7571

Table E2. ARI meta vs local HMMs

ticker	ari_meta_vs_local
AAPL	0.0256
INTC	0.1328
GOOG	0.6434
AMZN	0.1078
MSFT	0.1388

Table E3. Local MMD statistics (per ticker/metric)

ticker	metric	mmd_mean	toxicity_mean
AAPL	Price	0.9910	296566.6839
AAPL	OFI	0.2625	2663119121.6812
AAPL	OBI	0.2920	1.5184
INTC	Price	0.7573	21423.7190
INTC	OFI	1.2099	3198548572.2450
INTC	OBI	0.2920	2.0121
GOOG	Price	0.3152	1171971.7474
GOOG	OFI	1.4297	305362638.6396
GOOG	OBI	0.6062	1280588.4224
AMZN	Price	0.1543	1082727.2816
AMZN	OFI	0.9658	8829095883.7524
AMZN	OBI	1.3162	2882353.7475
MSFT	Price	0.1174	4682.1483
MSFT	OFI	0.2133	0.6893
MSFT	OBI	0.6340	1.4776

Appendix F: Implementation details

The complete analysis pipeline is implemented in Python using `polars` (v1.0+) for high-performance data loading, `scipy.stats` for Wasserstein distance computation and statistical tests, `hmmlearn` (v0.3) for Gaussian HMM estimation, `scikit-learn` (v1.3) for K-means clustering and

evaluation metrics, `numba` for optional JIT-compiled Wasserstein acceleration, and the Python `concurrent.futures` API for parallelised per-ticker optimization. Transfer Entropy is estimated via histogram-based discretization with $k = 2$ lags and 10 bins. MMD uses an RBF kernel with median heuristic bandwidth.

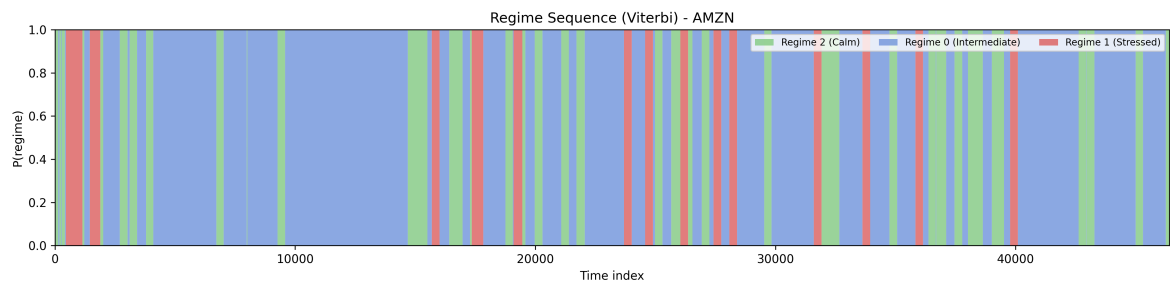
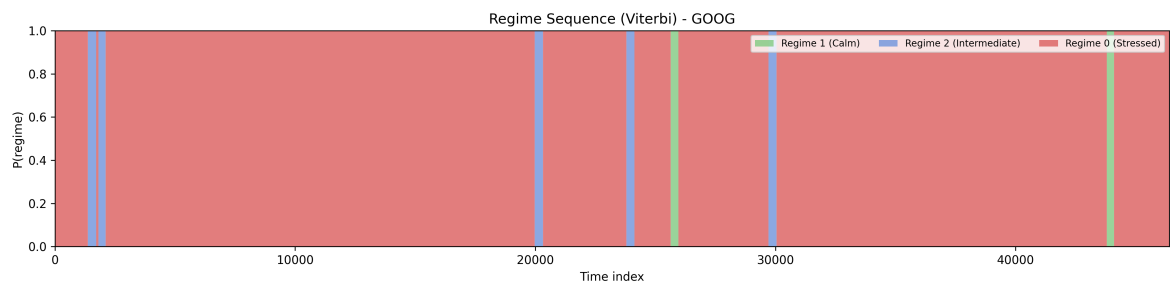
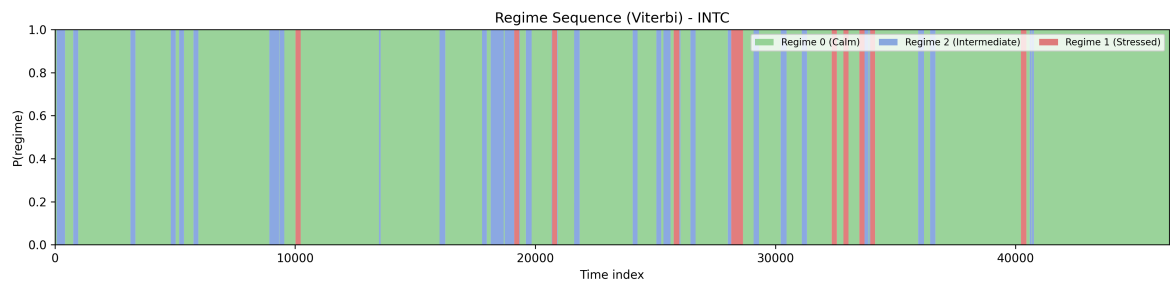
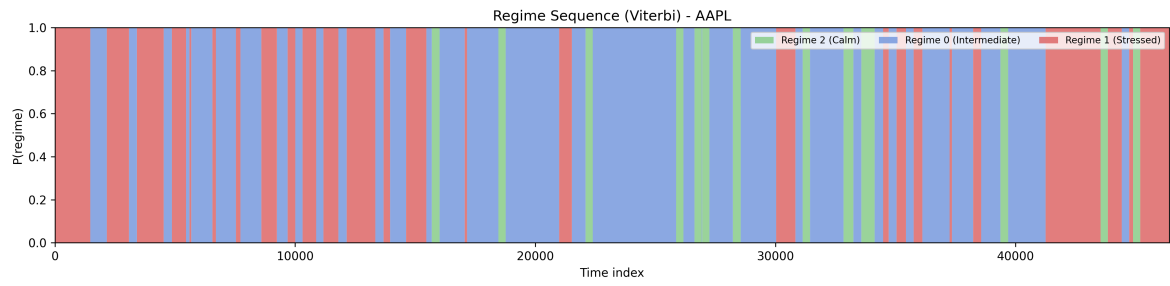
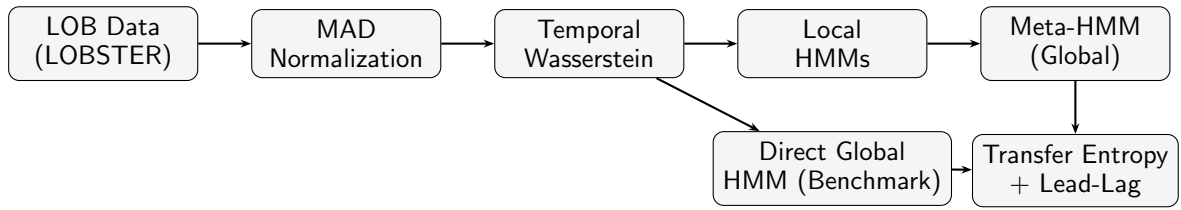
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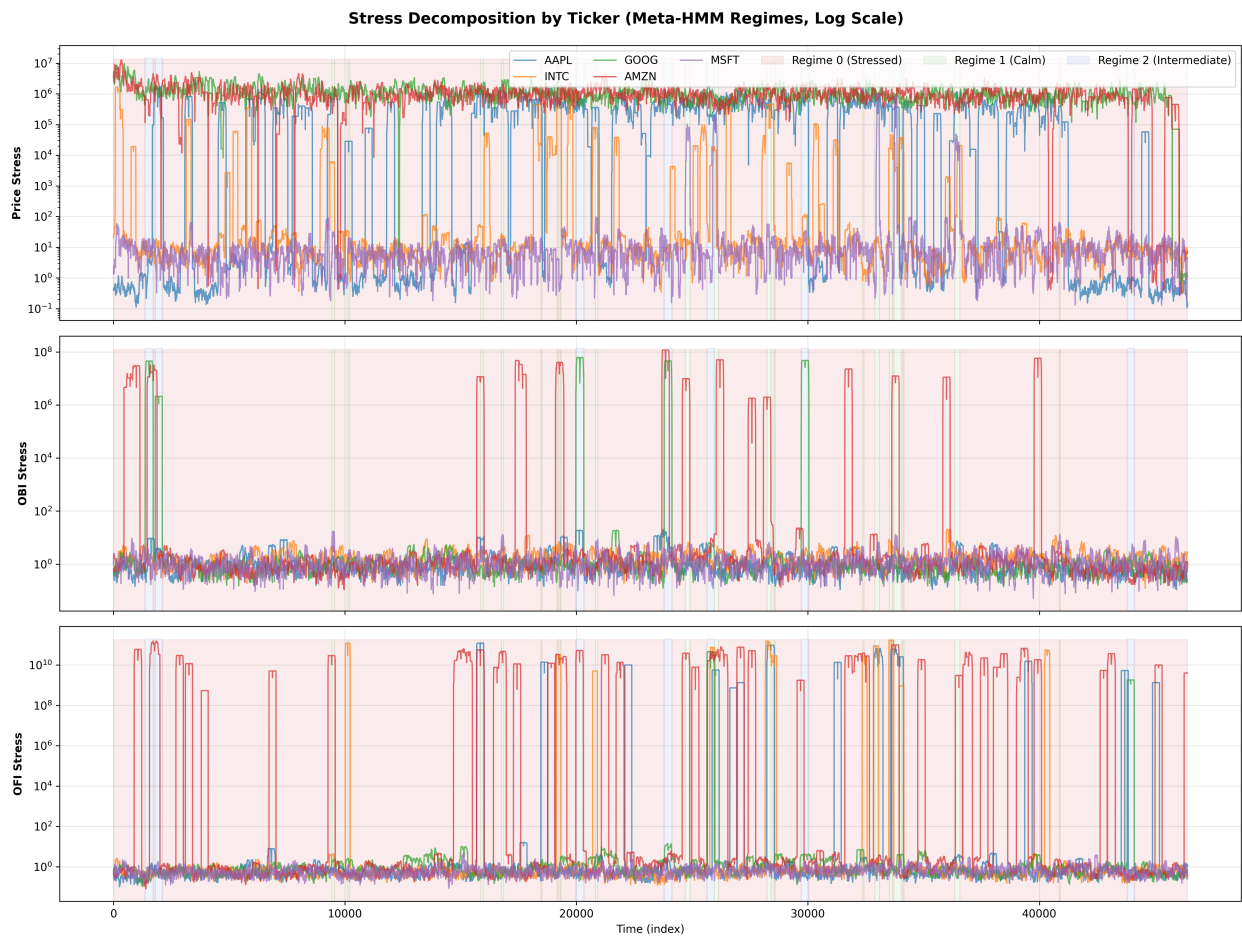
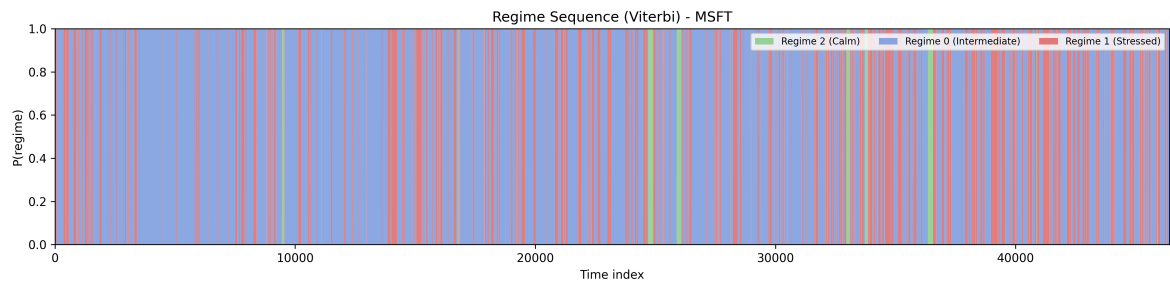
Table F1. This framework versus Horvath *et al.* (2021): four key extensions.

Dimension	Horvath et al. (2021)	This paper
Temporal resolution	Daily returns	500 ms LOB metrics
Normalization	Raw returns (no filtering)	MAD normalization (non-parametric)
Wasserstein type	Cross-sectional (between assets)	Temporal (before vs. after, within asset)
Temporal structure	K-means (no persistence)	Hierarchical HMM (Local + Meta)

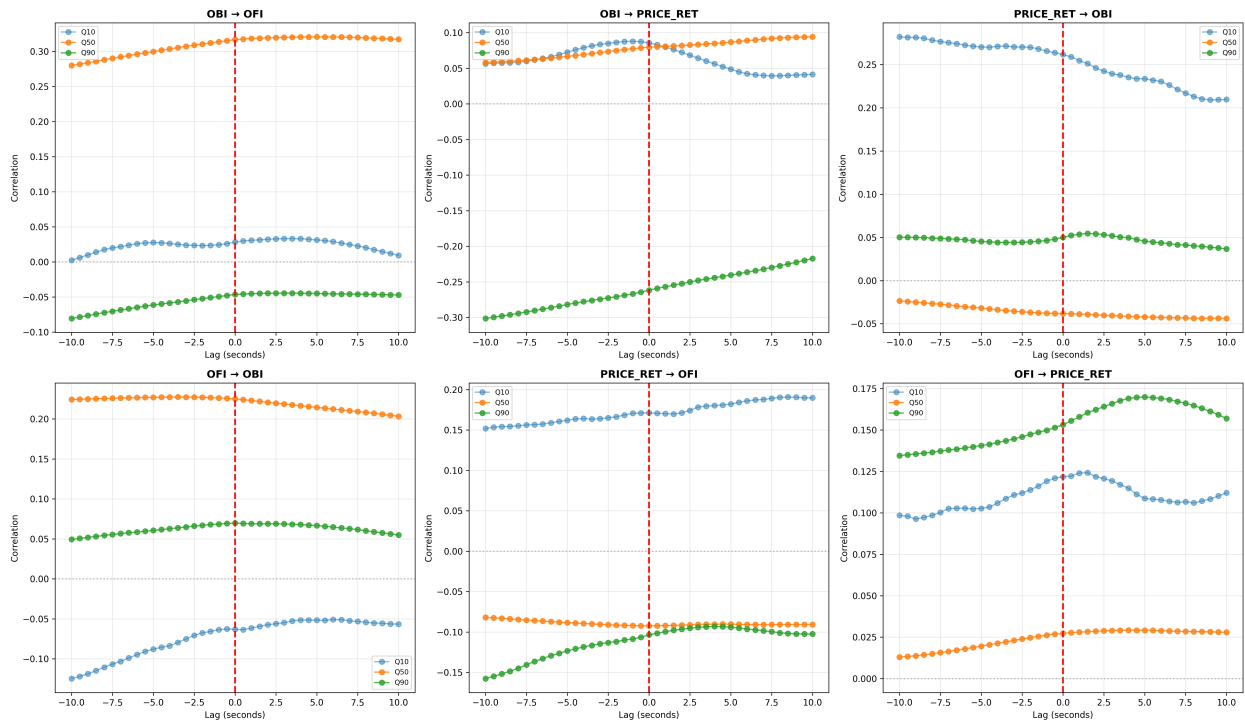
The temporal formulation (row 3) is the central innovation: it detects *when* a distributional change occurs within each asset, rather than *how different* two assets are, resolving the confound between persistent cross-asset differences and genuine regime transitions.

Figures

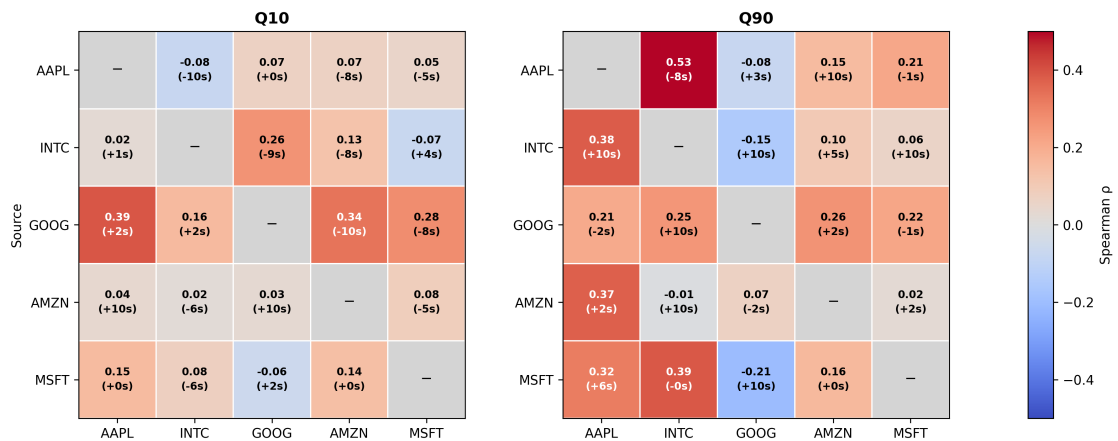




Multi-Metric Lead-Lag by Quantile — Sector (all tickers averaged)



Inter-ticker lead-lag: best correlation by stress quantile



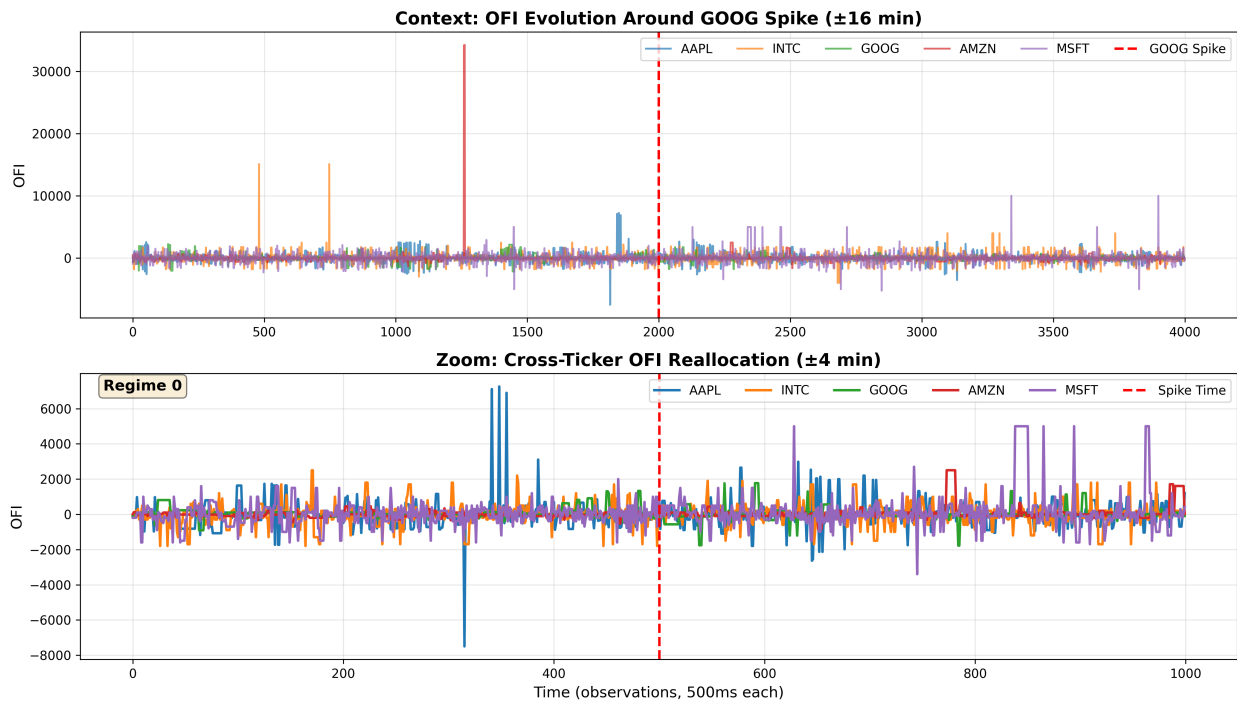


Figure Captions

Figure F. Analysis pipeline. LOB data are MAD-normalized per ticker; temporal Wasserstein distances (before vs. after each time point) serve as inputs to both per-ticker Local HMMs and a Direct Global HMM benchmark. Local posterior probabilities feed the Meta-HMM for sector-wide regime identification. Transfer Entropy and lead-lag analysis then characterize directed contagion.

Figure F. Viterbi regime sequence for AAPL (stacked area, one-hot encoded). Green = Calm (9.4%), blue = Intermediate (56.0%), red = Stressed (34.6%).

Figure F. Viterbi regime sequence for INTC. Green = Calm (80.6%), blue = Intermediate (14.2%), red = Stressed (5.2%).

Figure F. Viterbi regime sequence for GOOG. Green = Calm (1.3%), blue = Intermediate (3.6%), red = Stressed (95.1%). The dominant Stressed state carries the highest σ_r and $|\overline{\text{OBI}}|$ within GOOG's own intraday distribution; see Section 4 for interpretation.

Figure F. Viterbi regime sequence for AMZN. Green = Calm (23.2%), blue = Intermediate (65.6%), red = Stressed (11.2%). Stress ordering is driven by $|\overline{\text{OFI}}|$ (241 for Stressed) rather than return volatility.

Figure F. Viterbi regime sequence for MSFT. Green = Calm (2.4%), blue = Intermediate (67.4%), red = Stressed (30.2%).

Figure F. Wasserstein stress decomposition by ticker and metric with Meta-HMM regime overlay. Each row corresponds to one metric (Price, OBI, OFI); each colored line is one ticker. Colored backgrounds indicate the Meta-HMM global regime. Co-elevation of multiple tickers during stressed regimes validates the global identification.

Figure F. Multi-metric lead-lag analysis across stress quantiles (Q10, Q50, Q90). Each panel shows cross-correlations between source and target Wasserstein metrics. Negative lags indicate source precedes target. OFI autocorrelation exceeds 0.8 within ± 5 s; Q90 correlations substantially exceed Q10, confirming stress-driven amplification.

Figure F. Inter-ticker lead-lag correlation heatmap. Values represent the strongest Spearman correlation across quantiles for each ticker pair.

Figure F. Event study: GOOG OFI spike and cross-ticker co-movements (in-sample validation). *Top:* 33-minute context with the Wasserstein OFI spike visible at $t \approx 15,000$ ($\sim 11:33$ AM). *Bottom:* 4-minute zoom showing synchronized cross-ticker OFI responses. The episode is detected by $W_t^{(\text{GOOG}, \text{OFI})}$, occurs during an elevated-stress Meta-HMM regime, and confirms MSFT's Transfer Entropy-predicted role as the sector's primary stress absorber.